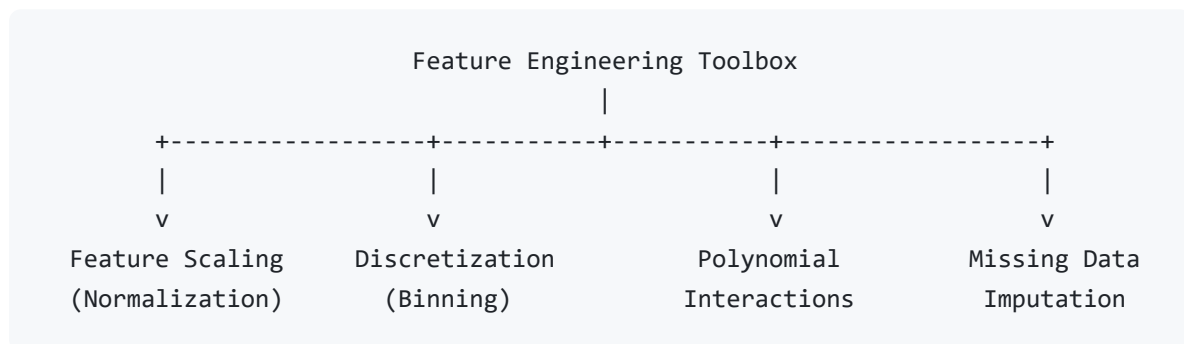


# Module 24.03: Other Feature Engineering Techniques

## 1. Taxonomy of Advanced Feature Transformations

Feature engineering is the process of transforming raw empirical attributes into informative representations that maximize the hypothesis capacity of machine learning algorithms:



## 2. Feature Scaling: Normalization vs. Standardization

When numerical features possess disparate scales ( $x_1 \in [0, 1]$  vs  $x_2 \in [10^4, 10^7]$ ), gradient descent oscillates violently and distance metrics are dominated entirely by large-magnitude variables.

### 1. Standardization (Z-Score Normalization)

Transforms distribution to zero mean ( $\mu = 0$ ) and unit standard deviation ( $\sigma = 1$ ):

$$z = \frac{x - \mu}{\sigma}$$

- **Properties:** Preserves outliers, unbounded  $z \in (-\infty, \infty)$ .
- **Best For:** Gradient Descent, Logistic/Linear Regression, PCA, Support Vector Machines.

## 2. Min-Max Normalization (Feature Scaling)

Compresses feature values into a rigid bounded interval, typically  $[0, 1]$ :

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

- **Properties:** Extremely sensitive to outliers (a single extreme value squashes remaining points into a narrow sliver).
- **Best For:** Neural networks with bounded activations, KNN, Image pixel processing ( $[0, 255] \rightarrow [0, 1]$ ).

## 3. Robust Scaling (IQR Scaling)

Removes the median and scales according to the Interquartile Range:

$$x_{\text{robust}} = \frac{x - Q_2}{Q_3 - Q_1} = \frac{x - \text{Median}}{\text{IQR}}$$

- **Best For:** Data laden with severe anomalies and heavy-tailed distributions.

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## 3. Discretization & Binning

Discretization converts continuous real features into discrete categorical bins.

- **Equal-Width Binning:** Divides the range  $[x_{\min}, x_{\max}]$  into  $K$  intervals of equal width:

$$w = \frac{x_{\max} - x_{\min}}{K}$$

- **Equal-Frequency (Quantile) Binning:** Partitions data such that each bin contains exactly  $\frac{m}{K}$  observations.
- **Utility:** Suppresses minor measurement noise and introduces non-linear step-function behavior into linear models.

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## 4. Polynomial Features & Interaction Terms

Linear models can capture complex non-linear curvatures through polynomial basis expansion:

$$\phi(x_1, x_2) = [1, x_1, x_2, x_1^2, x_1x_2, x_2^2]$$

- **Interaction Terms** ( $x_1x_2$ ): Encodes synergistic effects where the influence of feature  $x_1$  depends on the level of  $x_2$  (e.g., BMI  $\times$  Smoker).
- **Caution:** Feature count explodes combinatorially:  $\binom{d+k}{k}$  where  $d$  is feature count and  $k$  is polynomial degree.